TAKE IT FURTHER

Try this quick and sticky test. Be careful—the marshmallows can get very hot. You may want an adult to help out, just in case.

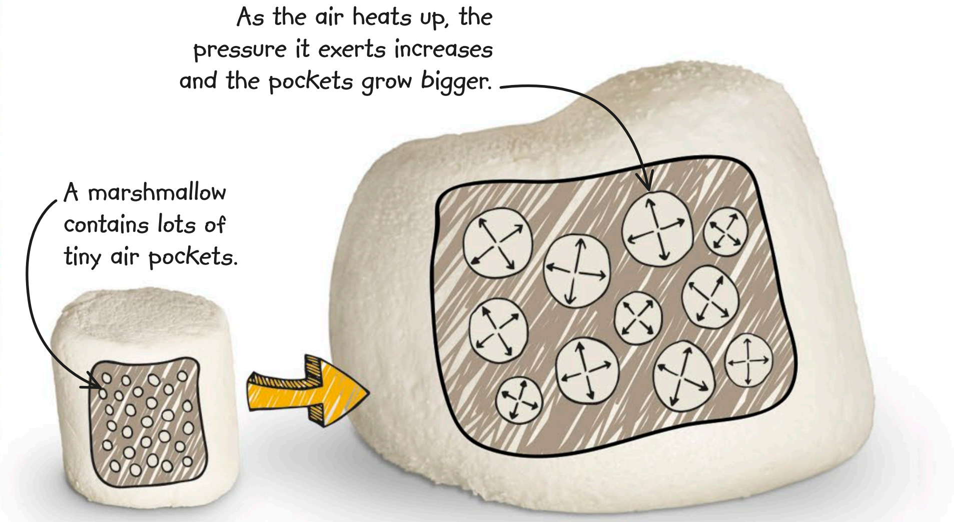
1 Grab a handful of mini marshmallows and assemble a little pyramid on a microwave-safe plate. Take some time perfecting your structure.

2 Set the microwave timer for a 30 second blast. Watch the marshmallows combine with each other, as they slump into a bubbling, pillowy mass.

3 Okay, are you ready to see something cool? Set the microwave timer for another 30 seconds. What have you got now? A plate of very sticky, hot liquid!

HOW IT WORKS

A marshmallow has a foamy, squashy texture because it has thousands of tiny air pockets inside it. Gases, such as air, are made of molecules moving freely at high speed and bouncing off any surfaces they meet. As they bounce, the molecules put pressure on the surfaces. Heating a gas causes the molecules to move faster, which increases the pressure. When you heat a marshmallow, each tiny air pocket blows up like a balloon.



The air bubbles inside an uncooked marshmallow are small and stable.

Very quickly, the air inside expands, pushing against the marshmallow's soft, sugary walls.

REAL WORLD SCIENCE
MELT IN THE MOUTH

A thickener called gelatin that is inside a marshmallow melts at a temperature just below that of the human body: 98.6°F (37°C). That is why marshmallows "melt in your mouth." The low melting temperature is important in your experiment because marshmallows become soft as they heat up, which makes it easier for them to expand. Another sweet that melts in the mouth is chocolate. The makers of chocolate carefully adjust their recipes to ensure that their products spread easily across your tongue.